Comparison of PID and Model Predictive Control (MPC) on a Heating System which has Thermal-Delay in sensors, Heat Loss and Gaussian Heater Output Variability

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Abstract

This study presents a comparison between Proportional-Integral-Derivative (PID) control and Model Predictive Control (MPC) for a residential heating system with a thermal model of a room. Controllers were implemented and optimised using differential evolution. Simulation results showed that MPC outperformed PID in both temperature stability and energy efficiency. The MPC's adaptive structure handled the delay and environmental disturbances better than the PID. These findings were interpreted through Ashby's concepts of ultrastability and biological homeostasis looking at MPC as a cybernetic controller capable of sustaining internal equilibrium in dynamic environments. In contrast, it can be inferred that predictive feedback strategies in modern home energy management systems can be highly valuable [5].

Keywords: Model Predictive Control (MPC), PID Control, Heating Systems, Thermal Delay, Heat Loss, Cybernetics, Homeostasis

1 Introduction

With growing interest in smart residential systems that maintains indoor thermal comfort despite unpredictable conditions the control of heating systems especially those that could handle thermal delays, heat loss, and non-linear heater responses while being energy efficient can be a complex challenge because it demands sophisticated techniques capable of real-time adaptation. The main questions is "MPC and PID control models, which one is more effective at reducing energy usage to improve sustainability in homes with typical insulation, heat retention, and delayed temperature feedback?". On one hand, PID control has been widely applied in industry due to its simplicity and robustness. There are also more advanced tuning techniques such as particle swarm optimization and firefly algorithm-based enhancements that improved PID's performance in systems with complex dynamics and time-delay characteristics [13, 10]. However, PID fundamentally operates as a reactive system which is unable to anticipate future states precisely for compensating the error.

On the other hand, MPC has become a powerful control in building energy systems. It predicts future system behavior by using an internal model and optimising control inputs

over a moving time horizon. Numerous studies have demonstrated the energy efficiency benefits and adaptability of MPC in heating applications. MPC showed a significantly better energy saving and comfort control when it is compared to rule-based and PID strategies in real environments [15, 19, 12].

An adaptive system is a system that can modify its behavior based on changes in the environment and in biological systems this is known as homeostasis. Ashby's concept of ultrastability describes systems that can change their internal structures to help them survival when the variables are outside safe bounds [2, 1]. MPC strategies have shown an incredible performance in systems with large thermal inertia and sensor delay. The recalibration of MPC is related to biological homeostasis like animals' ability to maintain their internal temperature stability through dynamic regulation and feedback [12]. Zhao et al. [19] proposed an MPC framework for radiant air systems with delay to optimise prediction intervals. They improved temperature stability of MPC over PID by 18.8% and saved energy. Similarly, Lv et al. [12] used adaptive MPC approach in a hybrid heating system and showed 22.2% energy reduction over PID. Cai et al. [5] worked on a learning-based MPC strategy for home energy management with reinforcement learning and their results showed improved performance compared to conventional ways. Joe and Karava [11] implemented MPC in radiant floor systems and had 34% cost savings compared to baseline control. Bianchini et al. [4] developed a demand-response MPC strategy that integrates economic signals in smart grid environments allowed intelligent participation with this strategy. Wang et al. [18] directly compared MPC and reinforcement learning (RL) and found that MPC outperformed RL in thermal control scenarios although they had limited training data and model uncertainty.

In this project, a residential room with thermal dynamics is simulated and affected by fixed sensor delays, constant-rate heat loss, and heater output variability. The aim is to determine which control method better maintains indoor thermal stability and minimises energy consumption, PID or MPC. This comparison is not handled simply an engineering task but viewed through a cybernetic's lens of how different controllers approximate the adaptive, self-regulating behaviors.

2 Methods

2.1 Thermal-Delay Heating Model

The indoor temperature $T(t)$ is changing by the first-order differential equation:

$$\frac{dT(t)}{dt} = \frac{1}{C}\left(Q_{\text{heater}}(t) - k(t)\cdot\left(T(t) - T_{\text{out}}(t)\right)\right) \tag{1}$$

where:

- C: Thermal capacitance of the system
- $Q_{\text{heater}}(t)$: Effective heater output power at time t
- $k(t)$: Time-varying heat loss coefficient
- $T(t)$: Indoor temperature
- $T_{\text{out}}(t)$: Outdoor (environmental) temperature

2.2 Heat-Loss Dynamics and Environmental Inputs

The heat loss coefficient is not constant and varies over time similarly implemented in adaptive control research for radiant systems [11, 12]. The outdoor temperature $T_{\text{out}}(t)$ is also not constant and has a 24-hour periodicity (1440 minutes) which is a typical day-night thermal variation:

$$T_{\text{out}}(t) = 10 + 4 \cdot \sin\left(\frac{2\pi t}{1440}\right) \tag{2}$$

$$k(t) = k_{\text{base}} + 0.1 \cdot \sin\left(\frac{2\pi t}{720}\right) \tag{3}$$

where $k_{\text{base}} = 4.0$. This models periodic changes in external weather influences such as wind or rain. The outside temperature is changing from 6°C to 14°C throughout the day.

2.3 Heater Model and Output Variability

The heater temperature is modeled in such way that reflects the gradual rise and fall of heat [19]. The internal heater temperature $T_h(t)$ is updated at each time step using the following equation:

$$T_h(t+1) = T_h(t) + \left(\frac{u(t)}{P_{\max}} \cdot \Delta T_{\max} - T_h(t)\right) \cdot \frac{\Delta t}{\tau} \tag{4}$$

where:

- $u(t)$ is the control signal (in watts) generated by the PID or MPC controller,
- $P_{\max} = 1500$ W is the maximum heater output,
- $\Delta T_{\max} = 50$°C is the maximum internal heater temperature,
- $\Delta t = 1$ minute is the simulation time step,
- $\tau = 5$ minutes is the heater time constant.

The heater's output power $Q_h(t)$ is calculated by scaling the internal heater temperature and adding zero-mean Gaussian noise:

$$Q_h(t) = \frac{T_h(t)}{\Delta T_{\max}} \cdot P_{\max} + \mathcal{N}(0, \sigma^2) \tag{5}$$

where $\sigma = 10$ W is the standard deviation of heater output noise [7]:

$$\mathcal{N}(0, \sigma^2) = \texttt{np.random.normal}(0, 10.0) \tag{6}$$

2.4 Sensor Delay

Sensor delay is implemented with a buffer storing past temperature values and represents latency in temperature measurements as discussed in previous papers [19, 18].

2.5 Simulation

Simulations were run over 1440 minutes (one day) with 1-minute resolution. Both controllers used the same room model and external conditions for fair comparison and noise was sampled in each time step. As mentioned before, the temperature and noise was different throught the day.

2.6 PID Controller Implementation

The PID controller is defined in the standard form:

$$u(t) = K_p e(t) + K_i \int e(t)dt + K_d \frac{de(t)}{dt} \tag{7}$$

where $e(t)$ is the error between the setpoint and delayed sensor reading. The (K_p, K_i, K_d) gains were optimized using differential evolution using optimization frameworks discussed in [13, 10, 17]. The error was used at each time step to update the control output with the proportional, integral, and derivative terms.

In practice, the derivative term K_D was set quite small because the sensor noise and delay can make a raw derivative term very unpredictable. A PI controller is also used in different scenarios for low-frequency control but with a touch of D it could improve its performance for these dampings. The PID gains (K_P, K_I, K_D) were tuned iteratively using a combination of methods.

First, the room's response to a heat step was approximated to get an idea of its gain and time constant and applied a Ziegler–Nichols style tuning on the delay-free part [3]. Then refined the gains through simulation trial-and-error to achieve a reasonable responsiveness and stability in the presence of delay. The final PID was stable and could hold temperature and it showed some overshoot for fast changes. A smoothness term weighted by the parameter $\Gamma = 0.6$ was introduced to penalise fast changes in heater power between succeeding time steps. This is because rapid fluctuations in heating output can lead to hardware fault, and unnecessary energy inefficiencies. By this term the controller was encouraged to produce more gradual and realistic control signals. The specific value of 0.6 was chosen through empirical tuning to balance temperature regulation.

2.7 MPC Controller Implementation

The MPC controller minimizes a cost function over a finite horizon $H = 60$ minutes:

$$J = \alpha \sum_{i=0}^{H} p(T_i, T_{\text{set}}) + \beta \sum_{i=0}^{H} u_i^2 \tag{8}$$

$$p(T_i, T_{\text{set}}) = \begin{cases} 2(T_i - T_{\text{set}})^2, & \text{if } T_i > T_{\text{set}} \\ (T_i - T_{\text{set}})^2, & \text{otherwise} \end{cases} \tag{9}$$

where H is the horizon length, u_i is the heater power at minute i, and $p(T_i, T_{\text{set}})$ is a penalty that penalizes overheating more than underheating. Optimization of $u_0, \ldots, u_H$ was solved using Scipy's `minimize` with bounds [17]. The design shows adaptive MPC formulas used in buildings [12], and its horizon control loop mimics homeostatic recalibration [5, 2, 1]. The parameters α and β which weigh temperature deviation and energy usage respectively, were optimized using differential evolution. The goal was to minimize the error plus a small energy penalty over the full simulation.

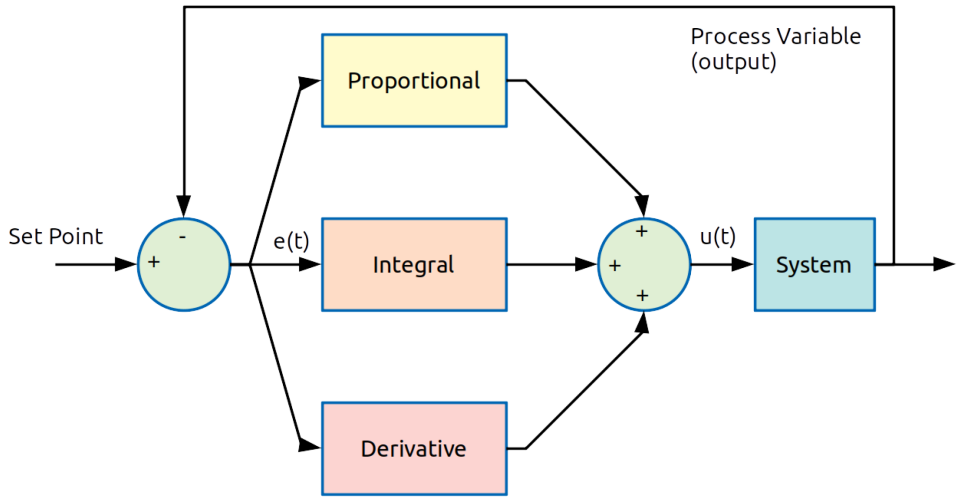

Figure 1: The diagram of the proportional–integral–derivative (PID) controller. The error function $e(t)$ is time dependent and it is computed as the difference between the set point and the process variable (system output). The controller then calculates further signal $u(t)$ by combining the proportional, integral, and derivative. The proportional term responds to current error, the integral term represents past errors, and the derivative term predicts future error [9].

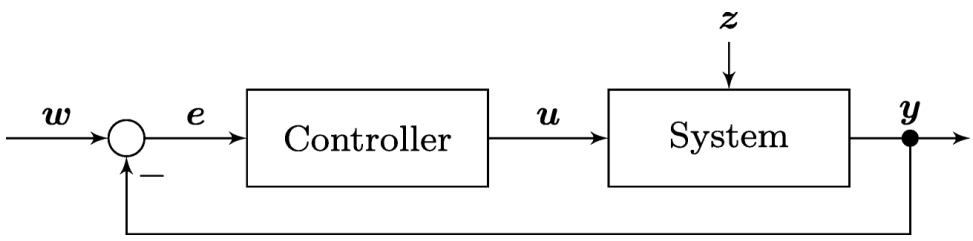

Figure 2: Standard feedback control loop with a controller like PID will have this shape. This configuration is a classical negative feedback loop, where the output is continuously compared with the setpoint to reduce error through feedback correction [14].

2.8 Visualization and Analysis

Temperature regulation and energy consumption were plotted to visually compare stability and efficiency [8]. RMSE from setpoint and cumulative energy (W) were primary metrics for evaluation strategies in [15, 4].

3 Results

3.1 Controller Optimization Outcomes

Multiple runs were used in simulation and variations across runs were observed. Differential evolution was used to optimize both control strategies. The best PID parameters were:

$$K_p = 86.05, \quad K_i = 0.16, \quad K_d = 4.2 \tag{10}$$

The best MPC cost weights were:

$$\alpha = 1100.0, \quad \beta = 0.103 \tag{11}$$

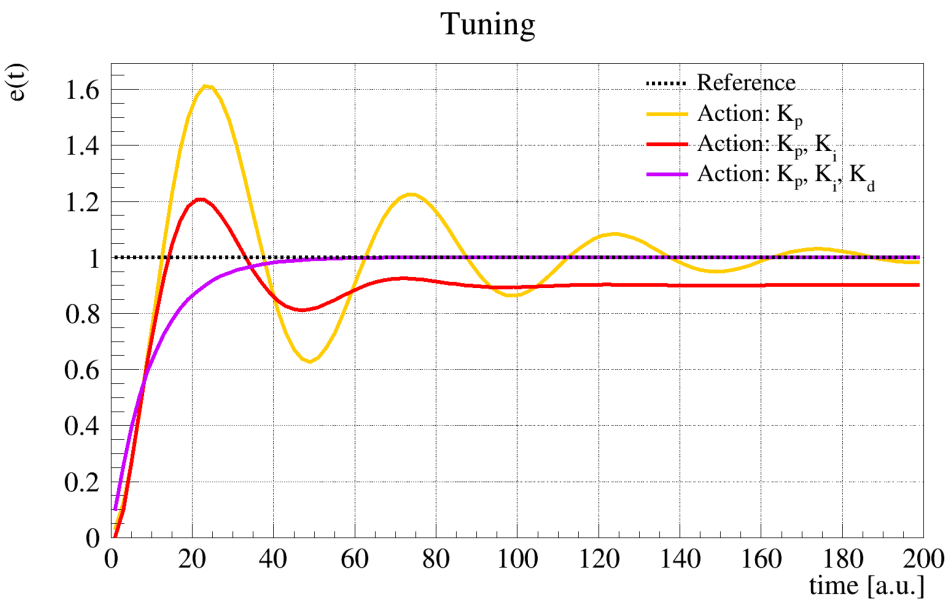

Figure 3: Simulation of a system's response to different PID controllers. The yellow curve is the result of pure proportional control (K_p), the red curve has the integral term (K_i), and the purple curve includes all three terms (K_p, K_i, K_d) [9]

These optimized values were used for the final performance comparison simulations.

3.2 Temperature Regulation Performance

The temperature response of both controllers over a 1440 minute (one day) period simulated with both methods and the MPC controller consistently maintained the room temperature closer to the setpoint of 22°C.

The root mean square error (RMSE) was:
From Time = $[0, 1440]$ (all simulation):

- **PID RMSE:** = 1.03 °C
- **MPC RMSE:** = 0.97 °C

From Time = $[60, 1440]$ (after room temp was normal):

- **PID RMSE:** = 0.12 °C
- **MPC RMSE:** = 0.09 °C

The average temp error was:
From Time = $[0, 1440]$ (all simulation):

- **PID Avg:** = 0.24 °C
- **MPC Avg:** = 0.21 °C

From Time = $[60, 1440]$ (after room temp was normal):

- **PID Avg:** = 0.10 °C

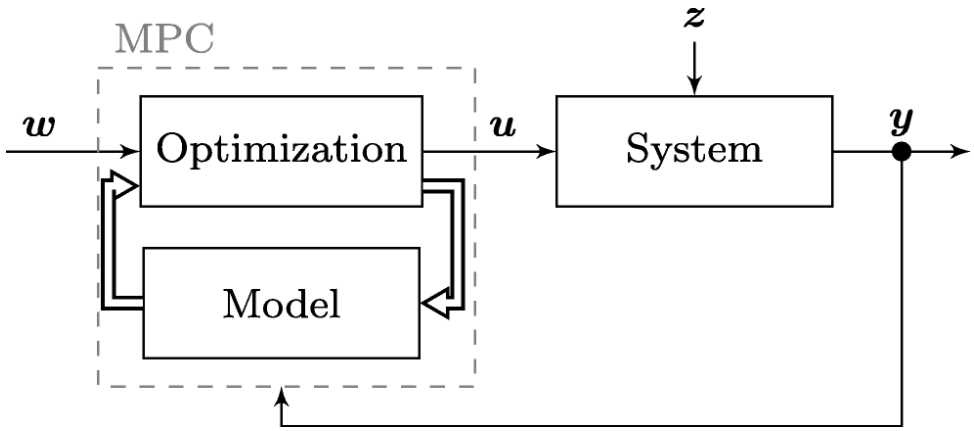

Figure 4: Model Predictive Control (MPC) system architecture. The MPC block contains a model of the system and an optimization that predicts future system behavior over a finite horizon. At each timestep the optimizer calculates a control input u that minimizes a cost function based on a reference trajectory w and predicted system outputs. The loop repeats with updated measurements and the structure forms a feedback loop and the output continuously used to update the internal model and re-optimise the future inputs [14].

- **MPC Avg:** $= 0.08\,^{\circ}\mathrm{C}$

The MPC controller showed stability under varying heat loss conditions and delayed sensor feedback. These results are consistent with findings in [19, 12, 11], and they all showed MPC performed better dynamic adaptation in thermal systems. When the temperature started from $6\,^{\circ}\mathrm{C}$ and the goal was $22\,^{\circ}\mathrm{C}$ both controllers used heater to raise the room temperature but due to the sensor delay the controllers did not see the temperature rise until a few minutes later and consumed a lot of energy. Then, oscillated a couple of times before settling around the setpoint. As the room warmed the MPC gradually narrowed the heating output.

3.3 Energy Consumption Comparison

Both controllers achieved thermal comfort but MPC strategy consumed less energy over the same time horizon.

- **Total Energy (PID):** $91614.03W$
- **Total Energy (MPC):** $91131.75W$

This was the energy usage after minute 30:

- **Total Energy (PID):** $70941.05W$
- **Total Energy (MPC):** $68945.41W$

There was a 0.52% reduction in energy usage overall and it was 2.81% reduction in energy usage after minute 30 and it supports the argument that MPC not only improves stability but also enhances energy efficiency which was also mentiond in prior experiments [15, 18, 5]. PID used $38.17kWh$ of heating and MPC $37.97kWh$ in a day of simulation. These findings align well with experimental results in Siroký et al. (2011) found 15–28%

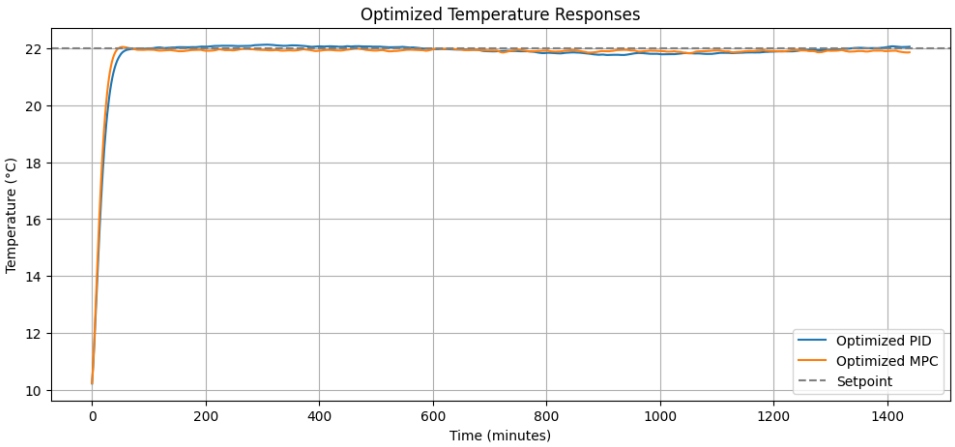

Figure 5: Optimised temperature responses of the PID and MPC control. Both controllers maintain the desired setpoint of 22°C after an initial warm-up. The MPC reaches the setpoint slightly faster and maintains a tighter bound around it. The PID occasionally overshoots and oscillates. The dashed black line is the temperature setpoint.

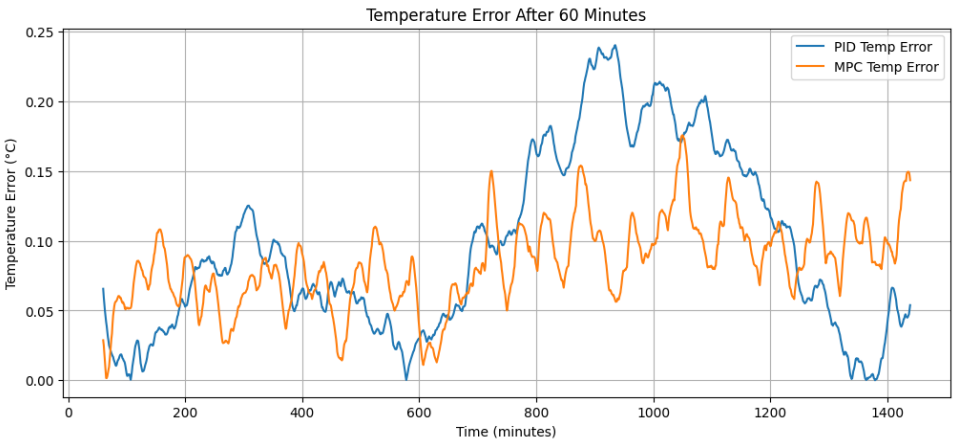

Figure 6: Temperature error for PID and MPC controllers after the first 60 minutes of operation. The PID controller shows more fluctuations and larger peak errors compared to the MPC. This shows the MPC's ability to better handle dynamic changes.

heating energy savings with MPC in a real buildings [15], and Joe & Karava (2019) reported around 16% savings for an MPC-controlled radiant system vs. conventional control [11]. Our scenario is relatively simple and our results has a narrower range. In more complex scenarios MPC could potentially save even more by optimizing around those factors.

3.4 Noise and Delay Compensation

Both controllers had identical Gaussian noise conditions in heater output and delayed temperature feedback. The MPC controller had less fluctuation and overshoot and it was more robust to variability that aligns with adaptive frameworks in previous works [2, 16].

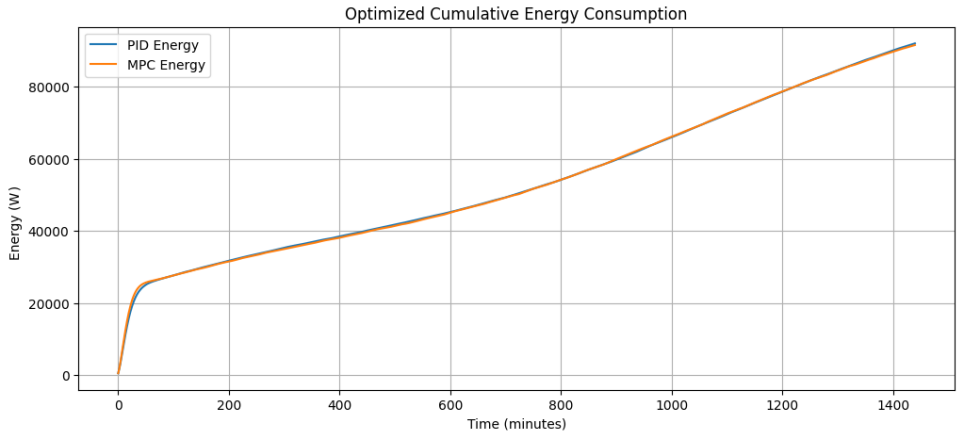

Figure 7: Optimized cumulative energy consumption of the PID and MPC controllers over the full operation period. After optimization the energy usage of both controllers becomes nearly identical in this plot but there is a figure below indicating that they have some differences when we zoom in.

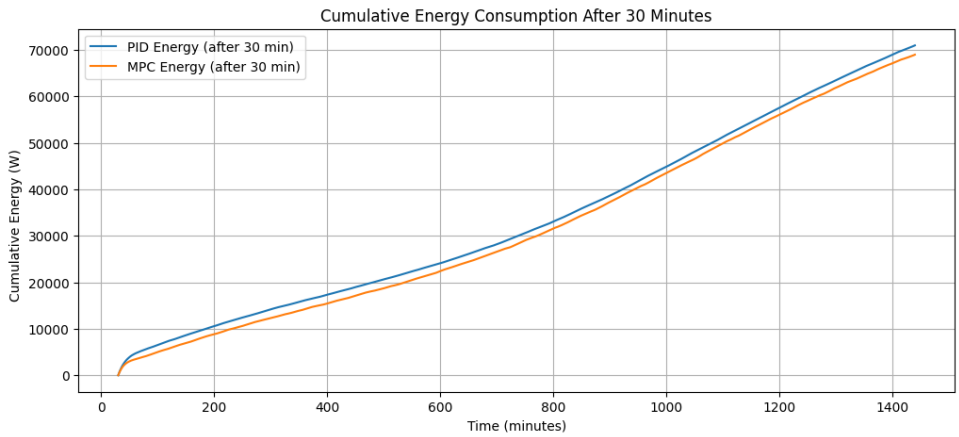

Figure 8: Cumulative energy consumption after the initial 30 minutes of control using PID and MPC strategies. The graph shows that the PID controller results in higher cumulative energy consumption and reflects the more aggressive or less predictive behavior of the PID.

4 Discussion

4.1 Implications of Performance Differences

The MPC controller outperformed the PID controller in both thermal stability and energy efficiency. The root mean square error (RMSE) from the temperature setpoint was 5.82% lower for MPC and energy usage was reduced by 0.52%. These findings endorse previous studies which shown MPC's advantage in managing thermal systems [15, 11, 12]. The improvement in temperature regulation of MPC is because it uses a predictive model to optimise heater outputs over a finite horizon. This approach allows it to compensate for delays and disturbances more effectively than the reactive PID controller. A PID controller can overcorrect or lag in response in case of sudden changes.

4.2 Effect of Sensor Delay and Heater Noise

Sensor delay and noise in heater output was challenging for both controllers. However, the simulation results showed that MPC's built-in model adaptation has robustness against these challenges and this is consistent with findings where MPC was specifically applied to delay-sensitive HVAC systems [19, 18]. The PID controller, despite being optimised using evolutionary techniques was more sensitive to noisy actuator behavior and sensor lag [13].

4.3 Ultrastability and Homeostasis

The concept of ultrastability introduced by Ashby can show us the differences between these controllers. A PID controller is a fixed feedback loop. It does not change its parameters automatically when challenged and a human would retune it if needed and in Ashby's terms PID is not ultrastable by itself although it is just stable within a range [2, 1]. The MPC changes its actions based on predictive knowledge and it is like an extra layer of feedback. While MPC in our case didn't rewrite its own algorithm, it did exhibit a higher form of stability by constantly recalculating its strategy [16]. The key difference is MPC's adjustments are computed using a model rather than random trial-and-error. In biological terms, MPC could be intrepreted as feed-forward physiological adjustments such as releasing adrenaline to raise heart rate in humans [6].

4.4 Energy Implications and Smart Home Relevance

The reduced energy consumption by MPC aligns with smart home goals of sustainability and cost efficiency as the importance of minimizing consumption without sacrificing comfort were discussed in prior experiments [5, 4].

4.5 Limitations and Future Work

One limitation of this study is the use of a simplified thermal model. The model was sufficient for simulation comparison but in real buildings extending this model is a neccessity in order to work and be applicable for future use.

Additionally, future research could explore hybrid control strategies that combine MPC's with PID, or investigate adaptive MPC methods that autonomously update model parameters over time [12, 5].

4.6 Implications of Reinforcement Learning

While this study focuses on comparing PID and MPC controllers the potential role of Reinforcement Learning (RL) in intelligent thermal control systems will always be an option for further study. RL offers a model-free approach where a controller learns optimal policies through interaction with the environment. Wang et al. [18] has shown that while RL-based controllers can match or exceed MPC in some cases, their performance is highly sensitive to training data, and reward structure design. Incorporating RL could enhance adaptability over longer timescales. Learning-based MPC strategies where RL is embedded to tune cost function weights has been explored by some research and it could be an option for hybrid usage of the models [5]. In conclusion, based on observed performance we can hypothesise that an adaptive MPC enhanced with online learning

like reinforcement learning could outperform both classical MPC and PID in dynamic environments and prior research supports this possibility [5].

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